

Calculus II Test 3 Study Guide

Test 3 will be on Monday March 30. It will cover the following sections/topics:

The percentages in parentheses tell approximately how much of the test will cover a particular topic.

✓ 7.8 Improper integrals (10%)

- Know the shortcut for determining if an integral of the form $\int_a^\infty \frac{1}{x^n} dx$ (where n is positive) converges or diverges. (If $n \leq 1$ it diverges; if $n > 1$ it converges.)
- Look at examples 1.0.1-1.0.3

✓ 8.1 Arc Length (25%, together with 8.2)

- Know the formulas for dS : $dS = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$ $dS = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$
- Know that Arc Length $= \int_a^b dS$
- Look at examples 1.0.2-2.0.1, ws8.2

✓ 8.2 Surface Area (25%, together with 8.1)

- Know that Surface Area $= 2\pi \int_a^b (\text{radius}) dS$
- Look at examples 0.0.1-0.0.4, w8.2

✓ 10.1 Parametric Curves

- This is all background information for 10.3. Be familiar with graphs like those mentioned in examples 1.0.2 and 1.0.3.
- Don't expect any problems that are specific to the material in this section.

10.2 Parametric Calculus (40%)

✓ Be able to use derivatives to describe the motion of a particle (left/right/up/down).

✓ Know that $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

- Know that $dS = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$
- Know how to do parametric arc length and surface area.
- Look at examples 1.0.1-2.0.2, 3.0.1-3.0.2, ws10.2

10.3 Nothing specifically from this section.

10.4 Areas in Polar Coordinates (25%)

- Know that the area inside $r = f(\theta)$ is $A = \frac{1}{2} \int_\alpha^\beta [f(\theta)]^2 d\theta$
- Know how many petals the rose curves $r = \sin(k\theta)$ and $r = \cos(k\theta)$ have.
- Be able to find the area of a petal of a rose curve or inside another polar graph.
- Look at examples 0.0.1, 0.0.2, ws10.4

7.8 Improper Integrals

Recall: We can interpret a definite integral as the area of a region.

So far, all of the definite integrals we've done involved regions with finite height and width.

Improper Integrals occur when either

- Type I: The region has infinite width:

$$\int_{-\infty}^b f(x) dx, \quad \int_a^{\infty} f(x) dx, \quad \text{or} \quad \int_{-\infty}^{\infty} f(x) dx$$

- Type II: The region has infinite height:

$$\int_a^b f(x) dx, \text{ where } f(x) \text{ has a vertical asymptote between } a \text{ and } b.$$

1 Type I: Infinite endpoint(s)

Fact: Let a be any positive number. Then the integral

$$\int_a^{\infty} \frac{1}{x^n} dx \quad \text{is} \quad \begin{cases} \text{infinite} & \text{if } n \leq 1 \\ \text{finite} & \text{if } n > 1 \end{cases}$$

Check it out here: <https://www.desmos.com/calculator/ztmfssfpfb>

These next three examples show why this rule holds true:

1.0.1 Example:

Evaluate the improper integral

$$\int_1^{\infty} \frac{1}{x^3} dx \quad \text{finite}$$
$$\int_1^{\infty} x^{-3} dx = \left. \frac{x^{-2}}{-2} \right|_1^{\infty} = -\frac{1}{2x^2} \Big|_1^{\infty} = 0 + \frac{1}{2} = \frac{1}{2}$$

1.0.2 Example:

Evaluate the improper integral

$$\int_1^{\infty} \frac{1}{x} dx \quad \text{infinite}$$

1.0.3 Example:

Evaluate the improper integral

$$\int_1^{\infty} \frac{1}{\sqrt{x}} dx \quad \text{infinite.}$$

1.1 Integrals from $-\infty$ to ∞

For integrals like this, you have to break them into two separate integrals, and do them independently:

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^b f(x) dx + \int_b^{\infty} f(x) dx$$

The break-point b can be any real number, so choose anything you want (0 is often convenient).

1.2 Comparisons

What about an integrals such as

$$\int_1^{\infty} \frac{1}{e^n} dx \quad \text{or} \quad \int_3^{\infty} \frac{1}{x^2 + x} dx$$

Memorize:

$$dS = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

The arc length of the curve from $x = a$ to $x = b$ is $\int_a^b dS$

1.0.1 Example:

Find the arc length of the function $f(x) = \frac{3}{5}x$ from $x = 0$ to $x = 10$.

$$\begin{aligned} dS &= \sqrt{1 + \left(\frac{3}{5}\right)^2} dx \\ &= \sqrt{1 + \frac{9}{25}} dx = \sqrt{\frac{34}{25}} dx = \frac{\sqrt{34}}{5} dx \end{aligned}$$

$$L = \int_0^{10} \frac{\sqrt{34}}{5} dx = \left[\frac{\sqrt{34}}{5} x \right]_0^{10} = \boxed{2\sqrt{34}}$$

1.0.2 Example:

Find the length of the parabola $y = x^2$ from $x = 0$ to $x = 3$.

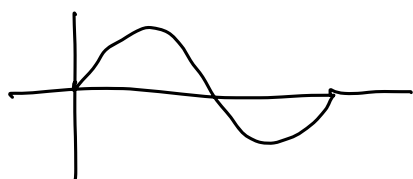
$$dS = \sqrt{1 + (2x)^2} dx$$

$$\int_0^3 \sqrt{1 + 4x^2} dx$$

8.1

1.0.3 Example:

Set up an integral for the length of one cycle of $y = \sin x$. Evaluate it using Desmos or another calculator.



$$dS = \sqrt{1 + (\cos x)^2} dx$$

$$\int_0^{2\pi} \sqrt{1 + \cos^2 x} dx$$

2 Length of $x = f(y)$

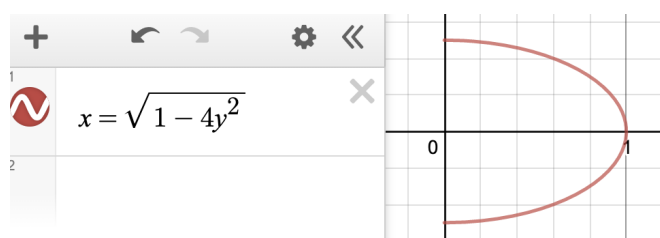
Memorize:

$$dS = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

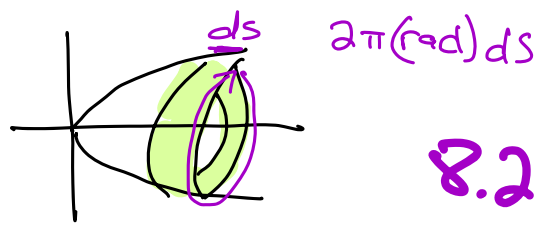
The arc length of the curve from $y = c$ to $y = d$ is $\int_c^d dS$

2.0.1 Example:

Set up an integral for the length of the half-ellipse shown here, and evaluate it using a calculator.



$$SA. = 2\pi \int (rad) dS$$



0.0.1 Example:

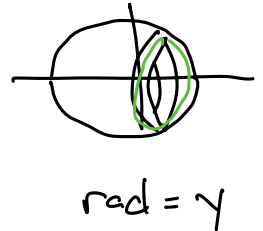
Find the surface area of the sphere formed by rotating the curve $y = \sqrt{4-x^2}$, from $x = -2$ to $x = 2$ around the x -axis.

$$dS = \sqrt{1 + \frac{x^2}{4-x^2}} dx$$

$$y' = \frac{-2x}{2\sqrt{4-x^2}} = \frac{-x}{\sqrt{4-x^2}}$$

$$S.A. = 2\pi \int_{-2}^2 \sqrt{4-x^2} \sqrt{1 + \frac{x^2}{4-x^2}} dx$$

$\frac{4-x^2}{4-x^2} + \frac{x^2}{4-x^2}$



$$= 2\pi \int_{-2}^2 \sqrt{4-x^2} \frac{\sqrt{4}}{\sqrt{4-x^2}} dx = 2\pi \int_{-2}^2 2 dx = 4\pi x \Big|_{-2}^2$$

$$= 4\pi(2) - 4\pi(-2) = \boxed{16\pi}$$

8.2

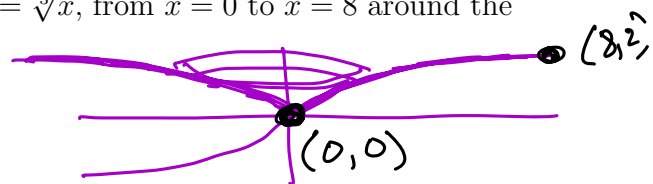
Note: You can use either form of dS when finding surface area:

$$dS = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{or} \quad \underline{dS = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy}$$

0.0.2 Example:

Find the area of the surface formed by rotating the curve $y = \sqrt[3]{x}$, from $x = 0$ to $x = 8$ around the y -axis. Do it both ways.

radius = x



$$y = x^{1/3}$$

$$y^3 = x$$

$$3y^2 = \frac{dx}{dy}$$

$$dS = \sqrt{1 + (3y^2)^2} \underline{dy}$$

$$S.A. = 2\pi \int_0^2 y^3 \sqrt{1 + 9y^4} dy$$

0.0.3 Example:

Rotate the curve $y = \sqrt{x-3}$, $3 \leq x \leq 19$ around the x -axis and find the surface area.

0.0.4 Example:

Rotate the curve $y = \ln x$, $1 \leq x \leq e^2$ around the x -axis and find the surface area. Set it up with respect to x and with respect to y , and evaluate whichever integral looks easier.

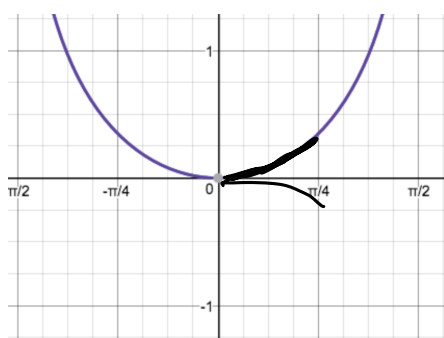
Worksheet 8.2

1. (a) Find the arclength of the function $y = \ln(\sec x)$ from $x = 0$ to $x = \frac{\pi}{4}$. Work the integral out by hand, and give both the exact value and a decimal approximation to three places.

- (b) For the same function, set up an integral for the surface area obtained by rotating the segment from $x = 0$ to $x = \frac{\pi}{4}$ around the x -axis. Use your calculator or a computer to get a decimal approximation to three places. (Don't try to do this one by hand).

- (c) Repeat part (b), but rotate around the y -axis.

$$y' = \frac{\sec x \tan x}{\sec x} = \tan x$$



$$dS = \sqrt{1 + (\tan x)^2} dx$$

$$dS = \sec x dx$$

$$A.L. = \int_0^{\pi/4} \sec x dx$$

b) $rad = y$ $S.A. = 2\pi \int_0^{\pi/4} \underbrace{\ln(\sec x)}_y \underbrace{\sec x dx}_{dS}$

c), $rad = x$ $S.A. = 2\pi \int_0^{\pi/4} x \sec x dx$

10.2 Parametric Calculus

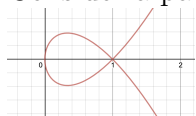
1 The Motion of a Function / Particle

It's helpful to think of parametric equations as describing the position of a particle. The main idea behind parametric calculus is that the motions of x and y don't depend (directly) on each other, but depend on time. Thus we are interested in the derivatives $\frac{dx}{dt}$ and $\frac{dy}{dt}$.

derivative	positive	negative	zero
$\frac{dx}{dt}$	right	left	stopped horizontally
$\frac{dy}{dt}$	up	down	stopped vertically

1.0.1 Example:

Consider a particle whose position is given by the equations $x = t^2$, $y = t^3 - t$.



left/right up/down.

When is the particle moving up/~~right~~? When $\frac{dy}{dt}$ is positive.

$$\frac{dy}{dt} = 3t^2 - 1 = 0$$

$$3t^2 = 1$$

$$t^2 = \frac{1}{3}$$

$$t = \pm \frac{1}{\sqrt{3}}$$



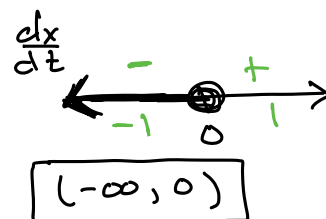
$$\left(-\infty, -\frac{1}{\sqrt{3}} \right) \cup \left(\frac{1}{\sqrt{3}}, \infty \right)$$

When is it moving left/~~right~~?

when $\frac{dx}{dt}$ is negative.

$$\frac{dx}{dt} = 2t = 0$$

$$t = 0$$



$$(-\infty, 0)$$

$$\text{slope} = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

1.1 Tangents

Notice that the curve in the previous example had both horizontal and vertical tangents.

Vertical tangents can occur when $\frac{dx}{dt} = 0$

Horizontal tangents can occur when $\frac{dy}{dt} = 0$

1.1.1 Example:

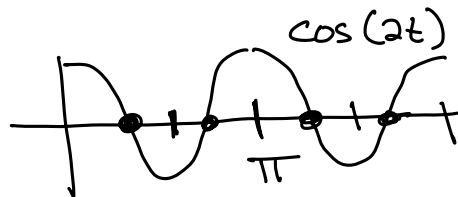
Consider a particle whose position is given by the equations $x = \sin(2t)$, $y = \cos(t)$ for $0 \leq t \leq 2\pi$. Determine when the particle is moving up, down, left, and right, and find the values of t where the graph has horizontal and vertical tangents.

Vertical tangent: when $\frac{dx}{dt} = 0$

$$\frac{dx}{dt} = 2\cos(2t) = 0$$

$$\cos(2t) = 0$$

$$\left\{ \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \right\}$$



1.2 Equations of Tangent Lines

Suppose you want to find the equation of a tangent line (in $y = mx + b$ form) at a given value of t . We need to know that

$$\text{slope} = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

1.2.1 Example:

Find the equation of the line tangent to the curve $x = t^2 - 2t$, $y = \sqrt{t}$ at the point where $t = 4$.

$$\frac{dx}{dt} = 2t - 2 \Rightarrow 2 \cdot 4 - 2 = 6$$

$$\frac{dy}{dt} = \frac{1}{2\sqrt{t}} \rightarrow \frac{1}{2\sqrt{4}} = \frac{1}{4}$$

$$\text{slope} = \frac{\frac{1}{4}}{6} = \frac{1}{24}$$

$$\text{point: } x = 4^2 - 2 \cdot 4 = 8$$

$$y = \sqrt{4} = 2$$

$$\text{Line: } y - 2 = \frac{1}{24}(x - 8)$$

2 Parametric Arc Length

We now use the parametric form of dS :

$$dS = \sqrt{\left(\frac{dy}{dt}\right)^2 + \left(\frac{dx}{dt}\right)^2} dt$$

We find arc length by using the formula

$$\text{distance traveled} = \int_{\alpha}^{\beta} dS.$$

If we want the length of the curve, we must ensure that **the curve is only traversed once** during the interval $t = \alpha$ to $t = \beta$. The next two examples show the difference in this:

2.0.1 Example:

Find the arc length, and the distance traveled by a particle with these parametric equations:

$$x = \sin t, \quad y = \cos t, \quad \underline{0 \leq t \leq 2\pi}$$

$$x' = \cos t \quad y' = -\sin t$$

$$dS = \sqrt{(\cos t)^2 + (\sin t)^2} dt$$

$$= dt$$

$$\boxed{A.L. = \int_0^{2\pi} dt}$$

2.0.2 Example:

Find the arc length, and the distance traveled by a particle with these parametric equations:

$$\underline{x = \sin(3t)}, \quad \underline{y = \cos(3t)}, \quad \underline{0 \leq t \leq 2\pi}$$

~~2.0.3 Example:~~~~Find the length of the curve $x = e^t + e^{-t}$, $y = 5 - 2t$, $0 \leq t \leq 3$~~ **3 Parametric Surface Area of Revolution****Recall:** Surface area $= 2\pi \int_{\alpha}^{\beta} (\text{radius}) dS$ Here we use the parametric dS , and we have to be sure that it's only traversed once.**3.0.1 Example:**The curve $x = 2 \sin t$, $y = 3 \cos t$, $0 \leq t \leq \pi$ is rotated around the y axis. Find the surface area.

$$x' = 2 \cos t \quad y' = -3 \sin t \quad \text{red} = x$$

$$dS = \sqrt{(2 \cos t)^2 + (-3 \sin t)^2} dt$$

$$S.A. = 2\pi \int_0^{\pi} \underbrace{2 \sin t}_{\text{radius}} \underbrace{\sqrt{4 \cos^2 t + 9 \sin^2 t} dt}_{dS}$$

3.0.2 Example:

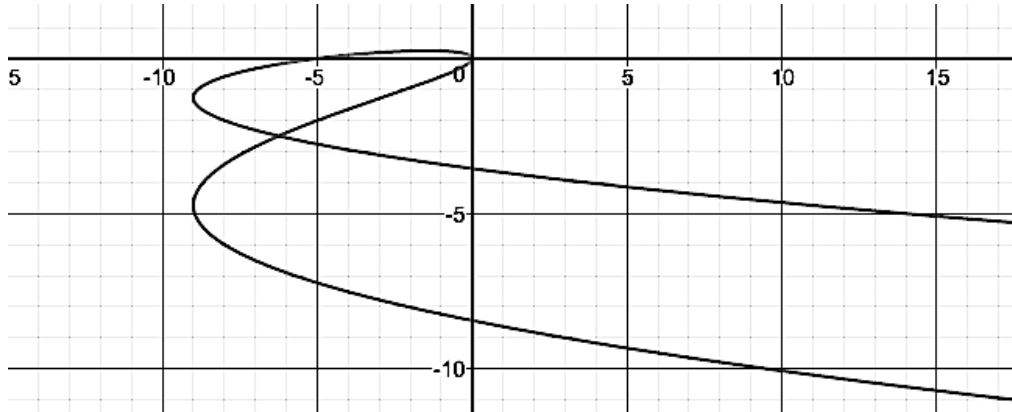
The curve $x = \underline{3t - t^3}$, $y = 3t^2$, $0 \leq t \leq 1$ is rotated around the x -axis. Find the surface area.

rad = γ

$$S.A. = 2\pi \int_0^1 3t^2 \sqrt{(3-3t^2)^2 + (6t)^2} dt$$

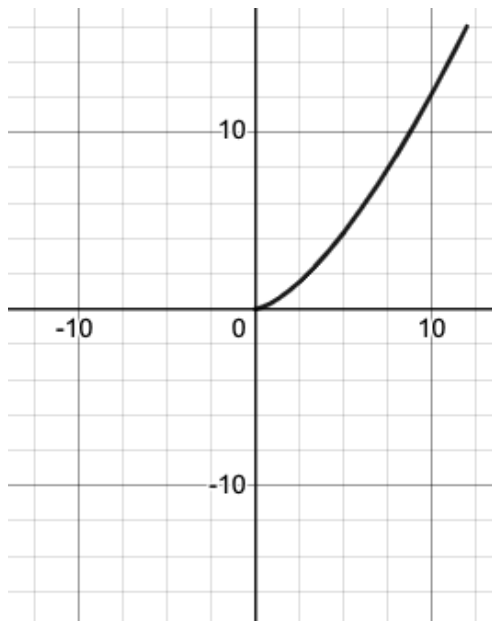
Worksheet 10.2

1. Here's the graph of the parametric curve $x = t^4 - 6t^2$ $y = t - t^2$



- (a) For which values of t is the function moving in each direction? Give exact answers using interval notation.
- i. up? ii. down? iii. left? iv. right?
- (b) Draw large dots on the graph at the points where it has horizontal and vertical tangent lines. Label the values of t at each.
- (c) Draw some arrows indicating the direction that a particle on the function would travel.
- (d) Find the equation of the tangent line at $t = 2$.

2. The parametric curve $x = 3t^2$, $y = 2t^3$ for $0 \leq t \leq 2$ is shown below.



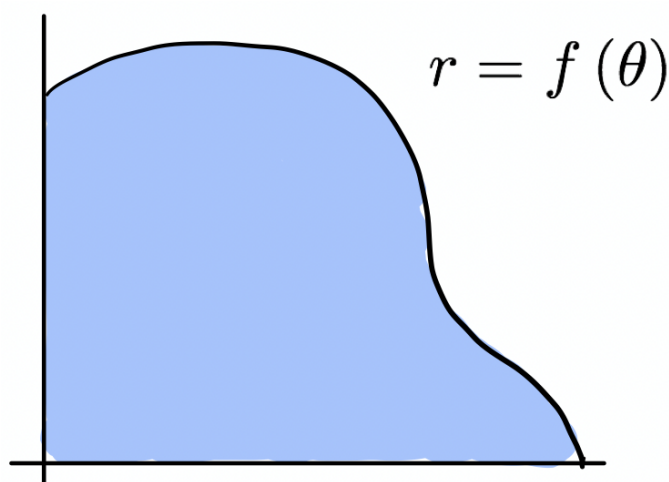
- (a) find the arclength Work the integral out by hand, and give both the exact value and a decimal approximation to three places.
- (b) For the same function, set up an integral for the area of the surface obtained by rotating the segment around the x -axis. Use your calculator or a computer to get a decimal approximation to three places. (Don't try to do this one by hand).
- (c) Repeat part (b), but rotate around the y -axis.
- (d) Repeat part (b), but rotate around the line $x = 12$.

10.4 Areas in Polar Coordinates

Recall:

- The area of a Circle with radius r is $A = \pi r^2$.
- The area of a sector of a circle with central angle θ (in radians) is $A_{\text{sector}} = \frac{1}{2}r^2\theta$

We use the sector area formula to find the area of an irregular curve:



Area inside $r = f(\theta)$:

$$A = \frac{1}{2} \int_{\alpha}^{\beta} [f(\theta)]^2 d\theta$$

Rose Curves: A function of the form $r = \sin(a\theta)$ or $r = \cos(a\theta)$ is called a **rose curve**. If a is even, the curve has $2a$ petals; if a is odd, it has a petals.

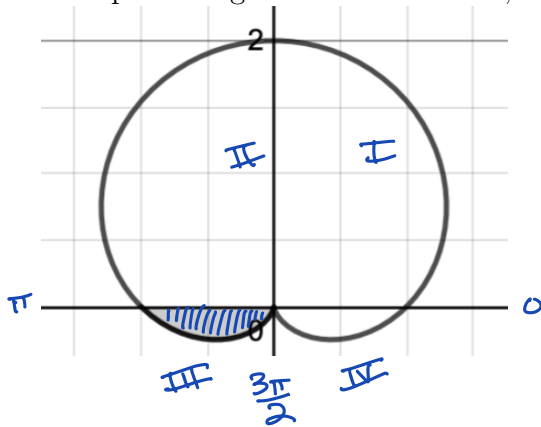
0.0.1 Example:

The rose curve $r = \sin 4\theta$ has 8 petals. Find the area of one petal:

$$\frac{1}{2} \int_0^{\pi/4} (\sin 4\theta)^2 d\theta$$

0.0.2 Example:

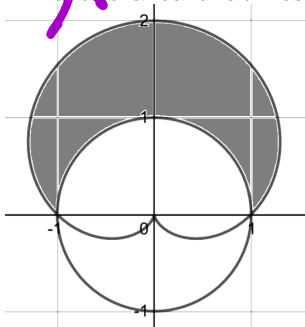
Set up an integral to find this area, and use a calculator to evaluate it: $r = 1 + \sin \theta$



$$A = \frac{1}{2} \int_{\pi}^{\frac{3\pi}{2}} (1 + \sin \theta)^2 d\theta$$

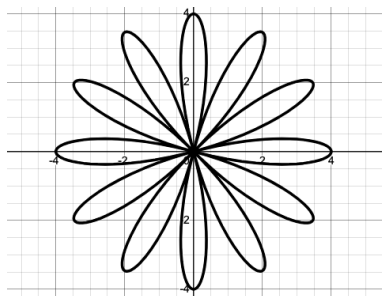
~~0.0.3~~ Example:

Find the area that lies outside of $r = 1$ and inside of $1 + \sin \theta$.



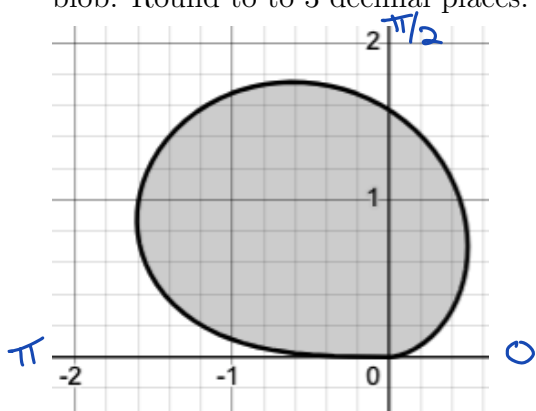
Worksheet 10.4: Two polar area problems.

- Find the area of one petal of the rose curve $r = 4 \cos(6\theta)$



$$\frac{1}{2} \int_0^{\pi/6} 16 \cos^2(6\theta) d\theta$$

- The blob below is the the graph of $r = \sqrt{\theta^2 \sin \theta}$ for $0 \leq \theta \leq \pi$. Find the area of the blob. Round to to 3 decimal places.



$$\frac{1}{2} \int_0^{\pi} \theta^2 \sin \theta d\theta$$

$$\begin{array}{l} + \theta^2 \quad \sin \theta \\ - 2\theta \quad - \cos \theta \\ + 2 \quad - \sin \theta \\ 0 \quad \cos \theta \end{array}$$

$$= \frac{1}{2} \left[-\theta^2 \cos \theta + 2\theta \sin \theta + 2 \cos \theta \right]_0^{\pi}$$

$$= \frac{1}{2} \left[-\pi^2(-1) + 2\pi(0) + 2(-1) \right] - \frac{1}{2} \left[2(1) \right]$$

$$\frac{\pi^2}{2} - 1 - 1 = \frac{\pi^2}{2} - 2$$